

MODULE 4 ASSIGNMENT

Solution 1 (Using File Operations)

```
rohit45@rohit45-VirtualBox:~$ cd Downloads/
```

```
rohit45@rohit45-VirtualBox:~/Downloads$ ls
```

```
input.txt
```

```
rohit45@rohit45-VirtualBox:~/Downloads$ cd
```

```
rohit45@rohit45-VirtualBox:~$ cat file_manip.sh
```

```
#!/bin/bash
```

```
substr=("frame.time" "wlan.fc.type" "wlan.fc.subtype")
```

```
>op_mod4.txt
```

```
while IFS= read -r line
```

```
do
```

```
    for ((i=0; i<3; i++))
```

```
    do
```

```
        if [[ "$line" == *"${substr[i]}"* ]]; then
```

```
            echo "$line" >> op_mod4.txt
```

```
            break
```

```
        fi
```

```
    done
```

```
done < "$1"
```

rohit45@rohit45-VirtualBox:~\$./file_manip.sh Downloads/input.txt &

[6] 21885

rohit45@rohit45-VirtualBox:~\$ cat op_mod4.txt

"frame.time": "Dec 31, 2024 12:50:25.890970000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

"frame.time": "Dec 31, 2024 12:50:25.892357000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

"frame.time": "Dec 31, 2024 12:50:25.928606000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

"frame.time": "Dec 31, 2024 12:50:25.983072000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

"frame.time": "Dec 31, 2024 12:50:25.983586000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

"frame.time": "Dec 31, 2024 12:50:25.993337000 India Standard Time",

"wlan.fc.type": "0",

"wlan.fc.subtype": "8",

rohit45@rohit45-VirtualBox:~\$

Solution 2 (Using Grep Command)

rohit45@rohit45-VirtualBox:~\$ cat file_manip1.sh

```
#!/bin/bash
```

```
grep -E '"frame.time"|"wlan.fc.type"|"wlan.fc.subtype"' "$1"
```

rohit45@rohit45-VirtualBox:~\$./file_manip1.sh Downloads/input.txt

```
"frame.time": "Dec 31, 2024 12:50:34.735026000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.789330000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.790306000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.799338000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.800782000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.837913000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.891718000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:34.892636000 India Standard Time",
```

```
"wlan.fc.type": "0",  
"wlan.fc.subtype": "8",  
"frame.time": "Dec 31, 2024 12:50:34.901914000 India Standard Time",  
"wlan.fc.type": "0",  
"wlan.fc.subtype": "8",  
"frame.time": "Dec 31, 2024 12:50:34.903035000 India Standard Time",  
"wlan.fc.type": "0",  
"wlan.fc.subtype": "8",  
"frame.time": "Dec 31, 2024 12:50:34.939826000 India Standard Time",  
"wlan.fc.type": "0",  
"wlan.fc.subtype": "8",
```

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rohit45@rohit45-VirtualBox:~\$

Solution 3 (Using AWK command)

rohit45@rohit45-VirtualBox:~\$ cat file_manip2.sh

```
#!/bin/bash
```

```
awk 'index($0,"frame.time") || index($0,"wlan.fc.type") || index($0,"wlan.fc.subtype")'  
"$1"
```

rohit45@rohit45-VirtualBox:~\$./file_manip2.sh Downloads/input.txt

```
"frame.time": "Dec 31, 2024 12:50:25.890970000 India Standard Time",  
"frame.time_utc": "Dec 31, 2024 07:20:25.890970000 UTC",  
"frame.time_epoch": "1735629625.890970000",  
"frame.time_delta": "0.000000000",  
"frame.time_delta_displayed": "0.000000000",  
"frame.time_relative": "0.000000000",  
"wlan.fc.type_subtype": "0x0008",  
"wlan.fc.type": "0",  
"wlan.fc.subtype": "8",  
"frame.time": "Dec 31, 2024 12:50:25.892357000 India Standard Time",  
"frame.time_utc": "Dec 31, 2024 07:20:25.892357000 UTC",  
"frame.time_epoch": "1735629625.892357000",
```

"frame.time_delta": "0.001387000",
"frame.time_delta_displayed": "0.001387000",
"frame.time_relative": "0.001387000",
"wlan.fc.type_subtype": "0x0008",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:25.928606000 India Standard Time",
"frame.time_utc": "Dec 31, 2024 07:20:25.928606000 UTC",
"frame.time_epoch": "1735629625.928606000",
"frame.time_delta": "0.036249000",
"frame.time_delta_displayed": "0.036249000",
"frame.time_relative": "0.037636000",
"wlan.fc.type_subtype": "0x0008",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",
"frame.time": "Dec 31, 2024 12:50:25.983072000 India Standard Time",
"frame.time_utc": "Dec 31, 2024 07:20:25.983072000 UTC",
"frame.time_epoch": "1735629625.983072000",
"frame.time_delta": "0.054466000",
"frame.time_delta_displayed": "0.054466000",
"frame.time_relative": "0.092102000",
"wlan.fc.type_subtype": "0x0008",
"wlan.fc.type": "0",
"wlan.fc.subtype": "8",

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rohit45@rohit45-VirtualBox:~\$